

Waqas Ahmad

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SUMMARY

Agricultural Engineer and Remote Sensing Researcher with 5+ years of experience in crop modeling, UAV sensing, and geospatial analytics. Skilled in Python, R, and integrating Earth observation data into process-based models for agricultural applications. Interested in digital twins, ecosystem fluxes, and sustainable farming systems.

RESEARCH EXPERIENCE

Engineer – Precision Agriculture | Apr 2024 – Feb 2025

Project: High-Throughput Crop Monitoring for Climate-Smart Agriculture (Funded by ADPC)

US-PAK Center for Advanced Studies in Agriculture & Food Security,

University of Agriculture Faisalabad (UAF), Pakistan

- Developed automated geospatial workflows in Google Earth Engine to derive field-scale crop indicators including canopy vigor, nitrogen demand, and water stress.
- Contributed to the development of satellite-based monitoring module for "Kisan360", a decision-support platform translating multi-sensor data into actionable insights for sustainable agroecosystem management.
- Assisted UAV-based field campaigns and stakeholder training for practical deployment of precision agriculture technologies.
- Led interdisciplinary technical reporting and capacity-building workshops for stakeholders, ensuring the alignment of satellite analytics with on-ground farmer requirements.

Research Associate – Precision Agriculture | Sep 2021 – Apr 2024

Project: Precision Agriculture & Analytics Lab (Funded by HEC)

US-PAK Center for Advanced Studies in Agriculture & Food Security,

University of Agriculture Faisalabad (UAF), Pakistan

- Developed a remote sensing data assimilation framework combining METRIC evapotranspiration outputs with the CropSyst process-based model for biomass and yield estimation in data-scarce regions.
- Participated in UAV field experiments using multispectral/thermal sensors, calibration workflows, GCP/RTK setup, and orthomosaic generation.
- Contributed to installation and field deployment of an eddy covariance flux tower for H₂O and CO₂ exchange monitoring.
- Contributed to multimodal data fusion experiments, integrating UAV multispectral imagery, Sentinel-1 SAR, and in-situ sensor data (LP-80 Ceptometer, SoilScan 360) for the physical validation of LAI and nitrogen stress models.
- Conducted climate, flood, and land-use analyses using Python, R, and Google Earth Engine.

PROFESSIONAL EXPERIENCE

Visiting Lecturer | Aug 2025 – Present

The Millennium Universal College, Faisalabad, Pakistan

- Instructing undergraduate courses in Programming and the Internet of Things, focusing on logic development, sensor integration, and data communication.
- Designing course materials, delivering lectures, and assessing student performance.

Self-Employed | 2014 – 2021

- Managed operations in family business, leading operations, basic finance, and client communications.

EDUCATION

- **M.Sc. (Hons) Agricultural Engineering | 2025**
University of Agriculture Faisalabad, Pakistan
CGPA: 3.65 / 4.0
Thesis: Satellite Data Assimilation into Mechanistic Crop Model for Wheat Yield Estimation
- **M.E. Electronics & Computer Engineering | 2014**
Griffith University, Australia
- **B.E Mechatronics Engineering | 2008**
Air University, Pakistan

CONFERENCE ABSTRACTS

- **Integration of Satellite Imagery and Cropping Systems Model to Determine Major Crop Yields in Data-Scarce Environments**
International Society of Photogrammetry and Remote Sensing Technical Symposium IV, Perth, Australia, 22-25 October 2024.
- **Mapping Flood Extent and Damage Assessment using Google Earth Engine**
International Conference on Geo-Informatics for Water and Agricultural Resource Management, University of Agriculture Faisalabad, Pakistan, 21-23 February 2024.

TECHNICAL SKILLS

- **Process-based Modeling & Data Assimilation**
 - **Mechanistic Modeling:** Expert in CropSyst for simulating crop growth, water balance, and yield; proficient in parameterizing rate and state variables for wheat cropping systems.
 - **Data Assimilation:** Developed automated frameworks to integrate satellite-derived biophysical parameters (ETa, LAI) into mechanistic engines to reduce simulation uncertainty.
 - **Energy & Light Balance:** Hands-on experience with METRIC/SEBAL energy balance models; foundational knowledge of Radiative Transfer Models (PROSAIL) and physical signal inversion.
- **Remote Sensing & Geospatial Analysis:**
 - **Satellite Platforms:** Multi-sensor expertise across Sentinel-1 & 2, MODIS & Landsat Satellites, and ERA5/TerraClimate (Climatology).

- **Geospatial Workflows:** Advanced time-series analysis, spatial gap-filling, temporal interpolation, and cloud-based image processing.
- **Software:** Proficiency in Google Earth Engine (GEE), QGIS, and ArcGIS.
- **Programming Languages:**
 - **R:** Advanced geospatial and statistical modeling using terra, sf, raster, and caret.
 - **Python:** Data science and machine learning pipelines using NumPy, Pandas, GeoPandas, rasterio, and scikit-learn.
 - **JavaScript:** GEE API development for high-throughput remote sensing analysis and visualization.
- **Field Instruments:**
 - **Flux Measurements:** Hands-on exposure to Eddy Covariance (Campbell Scientific IRGASON); knowledge of flux-based approaches for ET and atmospheric gas exchange.
 - **UAV Operations:** Flight planning and data processing for AgEagle eBee series and DJI Matrice series using Pix4D/WebODM for precision agriculture.
 - **Proximal Sensing:** Expert use of LP-80 Ceptometer (LAI validation), SoilScan360 (N/pH analysis), and various in-situ soil moisture probes.

ENGLISH LANGUAGE COMPETENCY

English: Professional Working Proficiency.

- **IELTS Academic:** Overall Band [7.5], Level: C1, Dated: 19 Apr 2026
 - **Listening:** 7.5
 - **Reading:** 8.5
 - **Writing:** 6.5
 - **Speaking:** 7.0

REFERENCES

- 1. Dr. Lubna Anjum** **(Academic Supervisor 1)**
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- 2. Prof. Dr. Saqib Ali** **(Academic Supervisor 2)**
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- 3. Dr. Azeem Khan** **(Professional Reference)**
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- 4. Mr. Salman Rizvi** **(Professional Reference)**
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